A Sharp Tail Barrier for Cucker–Smale Flocking

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Abstract

For the normalized Cucker–Smale flow we prove global existence, conserved mean velocity, exact variance dissipation, and a diameter comparison principle. The communication tail gives explicit confinement and exponential alignment, including unconditional flocking at the critical exponent $\beta = 1/2$. A scalar two-particle reduction proves the short-range threshold sharp and separates its zero-speed/unbounded-distance equality orbit from positive-speed escape. This round is the **global diameter-barrier owner**.

1 Model and all-particle barrier

For $N \geq 2$, $d \geq 1$, $K > 0$, and $\beta \geq 0$, set

$$\dot{x}_i = v_i, \qquad \dot{v}_i = \frac{K}{N}\sum_{j=1}^{N} \psi(|x_j - x_i|)(v_j - v_i), \qquad \psi(r) = (1 + r^2)^{-\beta}. \tag{1}$$

Write $\bar{v} = N^{-1}\sum_i v_i$ and $X = \max_{i,j} |x_i - x_j|$, $V = \max_{i,j} |v_i - v_j|$.

Theorem 1 (Global flow, dissipation, and tail barrier)**.** *Equation* (1) *has a unique global classical solution. The mean velocity is constant and*

$$\frac{d}{dt}\frac{1}{N}\sum_i |v_i - \bar{v}|^2 = -\frac{K}{N^2}\sum_{i,j} \psi(|x_i - x_j|)|v_i - v_j|^2. \tag{2}$$

The upper Dini derivatives satisfy

$$D^+X \leq V, \qquad D^+V \leq -K\psi(X)V. \tag{3}$$

If

$$V(0) < K\int_{X(0)}^{\infty} \psi(s)\, ds, \tag{4}$$

there is a unique $R \geq X(0)$ such that $K\int_{X(0)}^{R} \psi = V(0)$, and

$$X(t) \leq R, \qquad V(t) \leq V(0)e^{-K\psi(R)t}. \tag{5}$$

Every $x_i - x_j$ converges and every v_i converges to $\bar{v}$. Consequently every initial state flocks when $0 \leq \beta \leq 1/2$.

Proof. For any direction, the largest velocity projection cannot increase and the smallest cannot decrease; hence the velocity convex hull is invariant. Velocities remain bounded, positions grow at most linearly, and local existence extends globally. Pair symmetry gives $\sum_i \dot{v}_i = 0$ and, after pairing (i, j) with (j, i), gives (2) with its ordered-pair normalization.

If $V = 0$, all velocities coincide, every alignment acceleration vanishes, and the second inequality in (3) is immediate. Otherwise, at a velocity-diameter pair (p, q) put $e = (v_p - v_q)/V$ and $z_j = e \cdot v_j$. Then $z_q \leq z_j \leq z_p$. Every weight is at least $\psi(X)$, so

$$\dot{z}_p - \dot{z}_q \leq \frac{K\psi(X)}{N}\sum_j [(z_j - z_p) - (z_j - z_q)] = -K\psi(X)V.$$

Maximizing over active pairs proves the second Dini inequality; the identical active-pair argument for positions gives the first. Thus

$$D^+\left(V + K\int_{X(0)}^{X} \psi(s)\, ds\right) \leq 0.$$

The radius R supplied by (4) is a first-crossing barrier. Since $\psi(X) \geq \psi(R)$, Gronwall yields (5). Integrability of V makes relative positions Cauchy, and conservation identifies the common velocity. Finally $\psi(s) \sim s^{-2\beta}$, so its tail diverges precisely for $\beta \leq 1/2$. □